Elimination of the $q+2$ same-depth suffix channel

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For a positive odd integer x, put

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

For a positive exponent word $v = (a_1, \dots, a_\ell)$, define

$$\mathrm{B}(v) := \sum_{i=1}^{\ell} a_i, \qquad \mathrm{C}(v) := \sum_{i=1}^{\ell} 3^{\ell-i} 2^{a_1+\cdots+a_{i-1}}. \tag{2}$$

We use the record atoms

$$B := 12, \qquad C := 1113, \qquad D := 1122. \tag{3}$$

Their lengths, masses, and carries are

P	$\lvert P\rvert$	$\mathrm{B}(P)$	$\mathrm{C}(P)$
B	2	3	5
C	4	6	65
D	4	6	73.

(4)

1 The first-atom obstruction

Let $W = PU$, where $P \in \{B, C, D\}$, be an exact record-atom word from q to y. Compare it with an exact word V of the same raw length from z to y. At a common atom boundary write

$$\Delta := \mathrm{B}(V_{\mathrm{rem}}) - \mathrm{B}(W_{\mathrm{rem}}), \qquad z = 2^{\Delta} q + c, \qquad u := \frac{c}{2^{\Delta}}. \tag{5}$$

The arbitrary-prefix distortion theorem proves

$$\Delta \geq -1, \qquad -1 \leq u \leq 5. \tag{6}$$

The parity lattice gives even c when $\Delta = 0$ and odd c when $\Delta \geq 1$. If the remaining suffix is nonempty and $\Delta = -1$, the same theorem gives $c = -1/2$.

Suppose that the right state after P is (d, c), and that the alternative block aligned with P is a positive word v of length $\ell := |P|$ and mass s. The distortion transition is

$$\Delta_{\text{left}} = d + s - \mathrm{B}(P), \tag{7}$$

$$c_{\text{left}} = \frac{2^{\Delta_{\text{left}}}\,\mathrm{C}(P) + 2^s c - \mathrm{C}(v)}{3^\ell}. \tag{8}$$

Theorem 1 (the $q+2$ channel is empty)**.** *There is no exact same-depth word V from $q+2$ to y whose mass equals* $\mathrm{B}(W)$.

Proof. Such a word would give the left distortion state

$$(\Delta_{\text{left}}, c_{\text{left}}) = (0, 2). \tag{9}$$

Equations (7) and (8) would force

$$s = \mathrm{B}(P) - d, \qquad \mathrm{C}(v) = \mathrm{C}(P) + 2^s c - 2 \cdot 3^\ell. \tag{10}$$

Since $s \geq \ell$, the right defect satisfies

$$-1 \leq d \leq \mathrm{B}(P) - \ell. \tag{11}$$

Thus $d \in \{-1, 0, 1\}$ for $P = B$, and $d \in \{-1, 0, 1, 2\}$ for $P = C, D$.

The strip and parity lattice leave the following offsets:

d	c
-1	$-1/2$
0	$0, 2, 4$
1	$-1, 1, 3, 5, 7, 9$
2	$-3, -1, 1, \ldots, 19.$

(12)

For each row, the positive compositions of s into ℓ parts give the following carry sets:

P	d	$\{\mathrm{C}(v) : \lvert v\rvert = \ell,\ \mathrm{B}(v) = s\}$
B	-1	$\{5, 7, 11\}$
B	0	$\{5, 7\}$
B	1	$\{5\}$
C, D	-1	positive
C, D	0	$\{65, 73, 85, 89, 101, 103, 119, 125, 143, 179\}$
C, D	1	$\{65, 73, 85, 103\}$
C, D	2	$\{65\}$.

(13)

Substitution in the right side of (10) gives no member of the corresponding carry set. For $d = -1$, the required carry is negative. For $d = 0, 1$, its possible values are

P	d	required values from (10)
B	0	$-13, 3, 19$
B	1	$-17, -9, -1, 7, 15, 23$
C	0	$-97, 31, 159$
C	1	$-129, -65, -1, 63, 127, 191$
D	0	$-89, 39, 167$
D	1	$-121, -57, 7, 71, 135, 199.$

These sets are disjoint from the corresponding rows of (13). For $d = 2$, the required carry is congruent to 15 (mod 32) when $P = C$, and to 23 (mod 32) when $P = D$, while $65 \equiv 1 \pmod{32}$.

Hence the first atom cannot enter the state $(0, 2)$. This contradicts (9). □

Corollary 2 (sharpened same-depth suffix trichotomy)**.** *Let F be any exact word of length $|W|$ from a positive odd integer r to y. Then exactly one of the following holds:*

mass defect	*suffix start*
-1	$r = (q-1)/2, \quad P = C,$
0	$r = q, \quad F = W,$
$\delta \geq 1$	$r \geq 2^{\delta}(q-1).$

(14)

In particular, the formerly possible value $r = q + 2$ does not occur.

Proof. The arbitrary-prefix theorem gives the three rows in (14), except that its zero-defect row also allowed $q + 2$. Theorem 1 removes that value. The uniqueness assertion at $r = q$ is part of the same theorem. □

For the variable-depth inverse problem, every longer predecessor still splits into an inverse prefix and a same-depth suffix. Corollary 2 removes one complete nonminimal suffix channel. The deep prefix over the least suffix start, the actual q-suffix when it is nonminimal, and the positive-defect suffixes remain open.